Simply Typed Lambda Calculus

Dominik Kirst

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Slogans

- Imposes additional structure on computations
- Restricts to terminating computations
- Logical interpretation by Curry-Howard isomorphism
- Core of functional programming languages
- Extends to dependent type theories underlying proof assistants

Overview

Lecture 1:

- Simply typed lambda calculus
- Structural properties
- Normalisation
- Curry-Howard isomorphism

Lecture 2:

- Strong normalisation
- Extension with products

Lecture 3:

- Extension with sums
- Intersection types

Simple Types

Assuming some base types $A, B, C, \ldots$, we consider (simple) types:

- Every base type is a type.
- If σ and τ are types, then so is $\sigma \to \tau$.

In STLC, terms t are assigned types σ in contexts Γ, written $\Gamma \vdash t : \sigma$, as follows:

$$\frac{(x : \sigma) \in \Gamma}{\Gamma \vdash x : \sigma} \qquad \frac{\Gamma, x : \sigma \vdash t : \tau}{\Gamma \vdash \lambda x^{\sigma}.t : \sigma \to \tau} \qquad \frac{\Gamma \vdash t : \sigma \to \tau \quad \Gamma \vdash u : \sigma}{\Gamma \vdash tu : \tau}$$

In this presentation, variables x in abstractions $\lambda x^{\sigma}.t$ are usually annotated with types.

Example

$$\dfrac{\dfrac{\dfrac{\overline{\Gamma \vdash x : A \to B \to C} \quad \overline{\Gamma \vdash z : A}}{\Gamma \vdash xz : B \to C} \quad \dfrac{\overline{\Gamma \vdash y : A \to B} \quad \overline{\Gamma \vdash z : A}}{\Gamma \vdash yz : B}}{(x : A \to B \to C), (y : A \to B), (z : A) \vdash xz(yz) : C}}{\vdash \lambda xyz.xz(yz) : (A \to B \to C) \to (A \to B) \to A \to C}$$

Type Checking, Type Inference and Type Inhabitation

We distinguish three different decision problems related to typing:

1. Type checking: given Γ, t and σ, does $\Gamma \vdash t : \sigma$ hold?
   $\Rightarrow$ Decidable since the typing rules are deterministic!

2. Type inference: given Γ and t, is there σ such that $\Gamma \vdash t : \sigma$ holds?
   $\Rightarrow$ Decidable since most general types can be derived!

3. Type inhabitation: given σ, are there t and Γ such that $\Gamma \vdash t : \sigma$ holds?
   $\Rightarrow$ Decidable, but not trivially so!

In extensions of STLC, these may well become undecidable...

Weakening and Substitutivity

Fact (Weakening)

If $\Gamma \vdash t : \sigma$ *and* $\Gamma \subseteq \Delta$, *then* $\Delta \vdash t : \sigma$

Proof.

By induction on $\Gamma \vdash t : \sigma$ with Δ generalised. □

Fact (Substitutivity)

If $\Gamma, x : \tau \vdash t : \sigma$ *and* $\Gamma \vdash u : \tau$, *then* $\Gamma \vdash t\{u/x\} : \sigma$.

Proof.

By induction on $\Gamma, x : \tau \vdash t : \sigma$. □

Subject Reduction

Fact

If $\Gamma \vdash t : \sigma$ *and* $t \longrightarrow_\beta u$, *then* $\Gamma \vdash u : \sigma$.

Proof.

By induction on $t \longrightarrow_\beta u$ with Γ generalised:

- If $(\lambda x^\sigma.t)u \longrightarrow_\beta t\{u/x\}$ and $\Gamma \vdash (\lambda x^\sigma.t)u : \tau$, then we know that $\Gamma, x : \sigma \vdash t : \tau$ and $\Gamma \vdash u : \sigma$. Then the claim $\Gamma \vdash t\{u/x\} : \tau$ follows by substitutivity.
- If $(\lambda x^\sigma.t) \longrightarrow_\beta (\lambda x^\sigma.t')$ via $t \longrightarrow_\beta t'$ and $\Gamma \vdash (\lambda x^\sigma.t) : \sigma \to \tau$, then we know that $\Gamma, x : \sigma \vdash t : \tau$. Then by the IH for $\Gamma, x : \sigma$ we obtain $\Gamma, x : \sigma \vdash t' : \tau$ and thus $\Gamma \vdash (\lambda x^\sigma.t') : \sigma \to \tau$ as desired.
- If $tu \longrightarrow_\beta t'u$ via $t \longrightarrow_\beta t'$ and $\Gamma \vdash tu : \tau$, then we know that $\Gamma \vdash t : \sigma \to \tau$ and $\Gamma \vdash u : \sigma$. Then by the IH also $\Gamma \vdash t' : \sigma \to \tau$ and thus $\Gamma \vdash t'u : \tau$ as desired.
- If $tu \longrightarrow_\beta t'u$ via $u \longrightarrow_\beta u'$ and $\Gamma \vdash tu : \tau$, then we know that $\Gamma \vdash t : \sigma \to \tau$ and $\Gamma \vdash u : \sigma$. Then by the IH also $\Gamma \vdash u' : \sigma$ and thus $\Gamma \vdash tu' : \tau$ as desired. □

Subject Expansion

Is typing also preserved by expansion?

If $\Gamma \vdash u : \sigma$ and $t \longrightarrow_\beta u$, then $\Gamma \vdash t : \sigma$?

No!

$$(\lambda xy.x)(\lambda x.x)\Omega \;\longrightarrow_\beta^* \; \lambda x.x$$

Here $\lambda x.x$ on the left can be assigned any type $\sigma \to \sigma$ while Ω is untypable.

Weak Normalisation: Statement

For STLC, there are two notions of normalisation:

- *Weak normalisation* states that every well-typed term has a normal form.
- *Strong normalisation* states that every reduction sequence ends in a normal form.

The latter implies the former but the converse is not true in general...

In this lecture, we sketch a simple argument for the weaker property:

Theorem (Weak Normalisation)

If $\Gamma \vdash t : \sigma$, *then* t *has a normal form.*

Does the converse also hold? No, because xx is a normal form but untypable!

Weak Normalisation: Proof Sketch

Theorem (Weak Normalisation)

If $\Gamma \vdash t : \sigma$, *then* t *has a normal form.*

Proof Sketch.

We can define a well-order $o(t)$ such that whenever t is not yet normal, there is t' with $t \longrightarrow_\beta t'$ and $o(t') < o(t)$. Then the claim follows by well-foundedness of o.

- The degree of a type is its nesting depth of arrows.
- The degree of a redex is the degree of its arrow type.
- The degree $d(t)$ of a term t is the max degree of its redexes.
- Then $o(t) = (d(t), n)$ where and n is the number of redexes of max degree.

So if t is not yet normal, just reduce one of it's redexes of max degree to obtain t'. □

The proof of strong normalisation that we discuss next week will be more intricate...

The Magic Trick

Reconsider the typing example from the beginning:

$$\dfrac{\dfrac{\overline{\Gamma \vdash x : A \to B \to C} \quad \overline{\Gamma \vdash z : A}}{\Gamma \vdash xz : B \to C} \quad \dfrac{\overline{\Gamma \vdash y : A \to B} \quad \overline{\Gamma \vdash z : A}}{\Gamma \vdash yz : B}}{\dfrac{(x : A \to B \to C), (y : A \to B), (z : A) \vdash xz(yz) : C}{\vdash \lambda xyz.xz(yz) : (A \to B \to C) \to (A \to B) \to A \to C}}$$

What happens if we delete all term information?

The Magic Trick

Reconsider the typing example from the beginning:

$$\dfrac{\dfrac{\dfrac{\overline{\Gamma \vdash A \to B \to C} \quad \overline{\Gamma \vdash A}}{\Gamma \vdash B \to C} \quad \dfrac{\overline{\Gamma \vdash A \to B} \quad \overline{\Gamma \vdash A}}{\Gamma \vdash B}}{(A \to B \to C, A \to B, A) \vdash C}}{\vdash (A \to B \to C) \to (A \to B) \to A \to C}$$

We obtain a proof in (intuitionistic) natural deduction!

The Curry-Howard Isomorphism

We can interpret logical propositions as computational types!

- Proof checking becomes type checking
- Provability becomes type inhabitation
- Cut-elimination becomes normalisation

This isomorphism is most direct for intuitionistic logic but can also be extended to accommodate classical logic via well-known program transformations.

Perspective

This interpretation goes a long way:

Implication	Function types
Conjunction	Product types
Disjunction	Sum types
Truth	Unit type
Falsity	Void type
Universal quantification	Dependent product type
Existential quantification	Dependent sum type

This also tells us what the respective proofs are and how they compute!